Course Code	UCS3562	Course Name	USER EXPERIENCE DESIGN
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Experiment No. 8: Mobile UI Component Design System Using Figma

1 Objective

To design and organize the reusable components and interface elements for a mobile application while considering touch interaction, accessibility and platform conventions.

2 Tools Required

Figma

3 Methodology

Step 1: Selection of the Reference Application

The Swiggy food delivery app was chosen as the reference mobile application. It was selected because it has many repeated interface patterns (cards, lists, chips, steppers, bottom navigation), which makes it well suited for a reusable component library.

Step 2: Identify the Major Mobile Interaction Requirements

The following interaction requirements were identified:

i. **Tap:** Every action (add to cart, open restaurant, pay) must work with a single clear tap.

ii. **Scroll and swipe:** Restaurant lists scroll vertically and category or offer carousels swipe horizontally.

iii. **Text and voice search:** Users must find a dish or restaurant quickly using a prominent search field.

iv. **Location access:** The app must detect or let the user choose a delivery address in a few steps.

v. **Cart editing:** Quantity must be increased or decreased in place using a stepper without leaving the screen.

vi. **Secure checkout:** Payment, coupon and address selection must be simple, with clear error and success feedback.

vii. **Live order tracking:** The user must see order status and receive notifications in real time.

Step 3: Identify Commonly Used Interface Elements

The following interface elements were identified:

i. **Buttons:** Primary (filled), secondary (outlined) and text buttons, for example ADD, Log in, Proceed to pay.

ii. **Toggles and switches:** Veg-only switch, notification preferences, save address as default.

iii. **Checkboxes and radio buttons:** Add-ons selection and payment method selection.

iv. **Text fields:** Search bar, address and coupon code input with label, hint and error state.

v. **Chips:** Filters such as Pure Veg, Rating 4.0+, Fast delivery.

vi. **Cards:** Restaurant card and dish row with image, title, rating and price.

vii. **Steppers:** Quantity selector with minus, count and plus.

viii. **Badges and tags:** Bestseller, offer, rating and delivery time labels.

ix. **Navigation:** Bottom navigation bar, top app bar with back arrow and location selector, tabs.

x. **Overlays:** Bottom sheets for customisation and filters, dialogs and snackbars for feedback.

xi. **Progress indicators:** Order tracker steps and skeleton loaders.

Step 4: Establish Touch Target Guidelines

The touch target rules followed in the design system are given in Table **??**.

Guideline	Value used
Minimum touch target (iOS)	44 × 44 pt
Minimum touch target (Android)	48 × 48 dp
Value adopted in the design system	48 × 48 px minimum for every tappable element
Spacing between adjacent targets	At least 8 px
Thumb zone	Primary actions (ADD, View cart, Pay) placed in the lower half of the screen
Small visual, large hit area	Icons drawn at 24 px with padding to reach the 48 px target
Feedback	Pressed state change (darker fill or ripple) on every tappable element

Table 1: Touch target rules

Step 5: Apply Consistent Typography, Color Palette, Spacing and Iconography

5.1 Typography. The font family used is Figtree (Figma), with Helvetica/Arial as fallback. The type scale is given in Table **??**.

Style	Size / Line height	Weight	Used for
Heading (H1)	28 / 34 px	ExtraBold	Screen titles, hero text
Heading (H2)	20 / 26 px	Bold	Section titles
Subheading (H3)	16 / 22 px	Bold	Restaurant and dish names
Body	14 / 20 px	Medium	Descriptions, list text
Price	16 / 20 px	SemiBold	Item and bill amounts
Caption	12 / 16 px	Medium	Cuisine, distance, helper text
Button label	14 px	Bold	All buttons

Table 2: Type scale

5.2 Color palette. The colors used are listed in Table **??**. Text and background pairs were chosen to keep a contrast ratio of at least 4.5:1 for accessibility.

Swatch	Name	Hex	Usage
	Primary	#D9430B	Buttons, active tab, links
	Surface	#FFFFFF	Cards, sheets
	Black	#000000	Icons
	Text primary	#2B2B3A	Headings and body
	Text secondary	#6B6B76	Captions, hints
	Background	#F4F4F4	App background
	Highlight	#FFE9DC	Bestseller tag, selected chip
	Success / Veg	#1E8E4E	Veg mark, rating, success
	Error / Non-veg	#B3261E	Non-veg mark, errors
	Star	#E8A317	Rating star

Table 3: Color palette

5.3 Spacing and shape. Spacing scale: 4, 8, 12, 16, 24 px. Corner radius: 8 px for buttons and chips, 12 px for cards. Screen side margin: 16 px.

5.4 Iconography. A set of 20 line icons on a 24 px grid with 1.8 px stroke and rounded caps (home, search, cart, profile, favourite, location, rating, delivery time, notification, filter, back, offers, delivery, close, add, remove, wallet, orders, dropdown, support). Icon-only buttons carry text labels for screen readers.

Step 6: Apply Suitable Mobile Interaction Patterns

Bottom navigation for the main sections (Food, Search, Orders, Cart, Profile).

Pull to refresh on the restaurant feed.

Horizontal swipe carousels for categories and offers.

Bottom sheets for dish customisation, filters and address selection.

Sticky cart bar that appears after the first item is added.

In-place stepper that replaces the ADD button once an item is added.

Skeleton loaders while content loads instead of blank screens.

Immediate feedback through pressed states, snackbars and clear error messages.

Platform back gesture and back arrow supported on every inner screen.

Step 7: Organize the Components into a Reusable Figma Library

i. **Pages:** Cover, Typography, Color, Icons, Components, Patterns, Screens.

ii. **Styles and variables:** Text styles, color styles, spacing and radius stored as variables.

iii. **Components with variants:** Each component has variants for state (default, pressed, disabled, error) and size.

iv. **Auto layout:** Used so that components resize with content and keep spacing consistent.

v. **Naming convention:** Category/Component/Variant, for example Button/Primary/Default.

vi. **Nesting:** Larger components (restaurant card, bottom nav) are built from smaller ones (tags, icons, buttons).

vii. **Publishing:** The file is published as a team library so that every screen uses the same components and updates stay in sync.

4 Screenshots of the Work

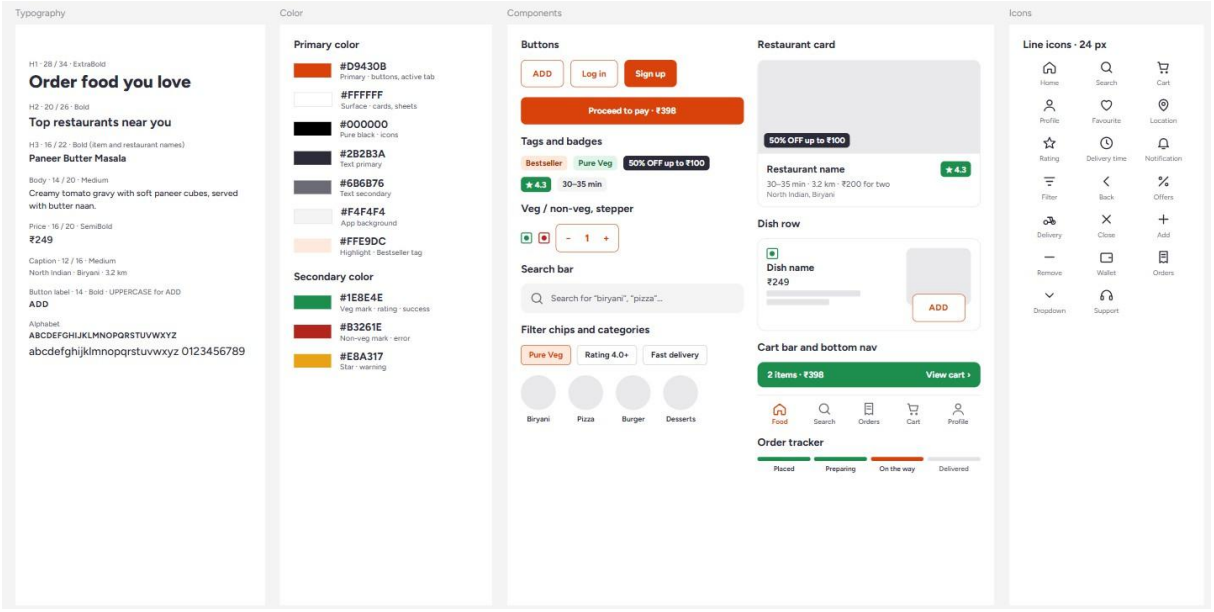


Figure 1: Entire Design System (Typography, Color Palette, Components and Icons)

5 Learning Outcomes

1. Designed reusable mobile components and applied mobile interaction principles.
2. Maintained consistency across mobile screens.

6 Result

The reusable components and interface elements of a food delivery mobile application, with the Swiggy app as reference, were designed and organized in Figma. Touch target

guidelines, typography, color palette, spacing and iconography were applied consistently, and the components were arranged into a reusable library.